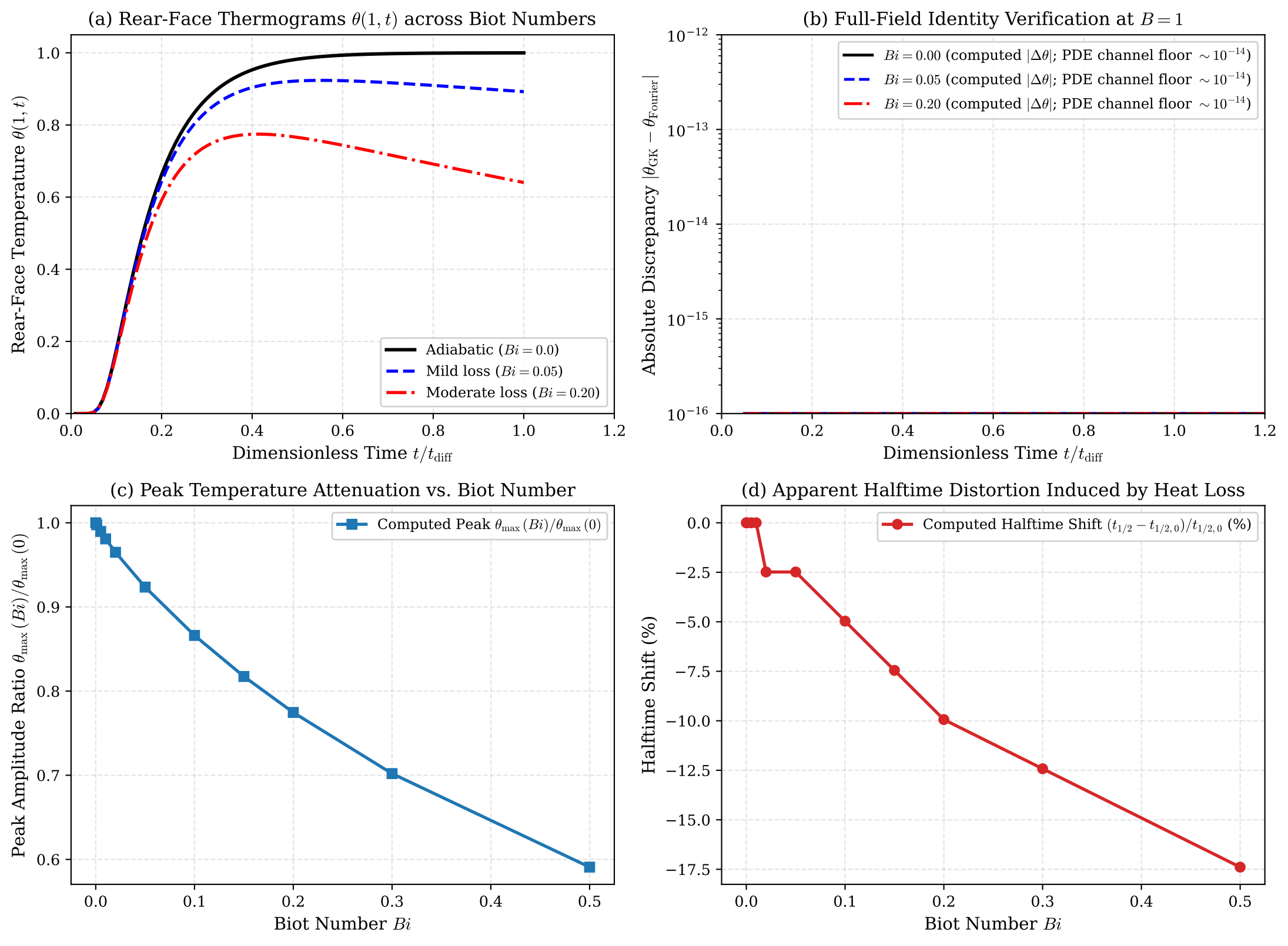
(a) Rear-Face Thermograms $\theta(1,t)$ across Biot Numbers
Rear-Face Temperature $\theta(1,t)$
Dimensionless Time t/t_{diff}
Adiabatic ($Bi=0.0$)
Mild loss ($Bi=0.05$)
Moderate loss ($Bi=0.20$)
(b) Full-Field Identity Verification at $B=1$
Absolute Discrepancy $|\theta_{\mathrm{GK}}-\theta_{\mathrm{Fourier}}|$
Dimensionless Time t/t_{diff}
$Bi=0.00$ (computed $|\Delta\theta|$; PDE channel floor $\sim 10^{-14}$)
$Bi=0.05$ (computed $|\Delta\theta|$; PDE channel floor $\sim 10^{-14}$)
$Bi=0.20$ (computed $|\Delta\theta|$; PDE channel floor $\sim 10^{-14}$)
(c) Peak Temperature Attenuation vs. Biot Number
Peak Amplitude Ratio $\theta_{\max}(Bi)/\theta_{\max}(0)$
Biot Number Bi
Computed Peak $\theta_{\max}(Bi)/\theta_{\max}(0)$
(d) Apparent Halftime Distortion Induced by Heat Loss
Halftime Shift (%)
Biot Number Bi
Computed Halftime Shift $(t_{1/2}-t_{1/2,0})/t_{1/2,0}$ (%)